What about data that's spread across multiple tables? This is where FOREIGN KEY constraints can help you keep data relationships intact.

Enforcing Relationships With Foreign Keys

The paper *Feral Concurrency Control: An Empirical Investigation of Modern Application Integrity*[3] covers how Active Record models are linked together using Associations and Validations within the Ruby application code layer.

Active Record uses association methods like has_many and belongs_to to express relationship types like one to one, and one to many. Validations are added to models to enforce validation that there aren't NULL values, and for many other things.

When a database table row depends on the existence of a row in another table, a column can be used to store a reference to the Primary Key id value of the other table being referred to.

That column can optionally have a Foreign Key constraint added to it. Trying to delete either row when a Foreign Key constraint is defined will be prevented by PostgreSQL.

Foreign Key constraints are not added by default so it's up to developers to add them.

For many years Rails did not support database Foreign Key constraints. This might be why they had less usage by Rails developers. Although support was added with a gem, native support for database Foreign Keys was added to Rails Version 4.2 released in 2014.

As a refresher, let's review FOREIGN KEY constraints[4] in the Rideshare database.

A Rider creates a Trip Request. When a Driver accepts a Trip Request a Trip record is created. Trips are linked to Trip Requests.

Since a Trip depends on a Trip Request, the trip_request_id is a Foreign Key column and has a NOT NULL constraint and Foreign Key constraint. Trip Requests are also linked with Riders, and Trips are linked with Drivers.

The Migration in Ruby turns into a SQL statement similar to the ALTER TABLE DDL SQL below. You don't need to run this statement since this Foreign Key constraint already exists in Rideshare.

3. http://www.bailis.org/papers/feral-sigmod2015.pdf
4. https://guides.rubyonrails.org/active_record_migrations.html#foreign-keys